

Ramon Garcia-Figueroa

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Professional Summary

Electrical/Electronic Engineering Technology with hands-on experience in industrial laser systems, power electronics, PCB testing, troubleshooting, automation, and system integration. Experienced with MATLAB/Octave, Python, PLC programming, electrical measurements, and engineering documentation. Strong background in electrical testing, production support, and hardware development.

Technical Skills

Power Electronics, Circuit Analysis, Electrical Troubleshooting, PCB Testing, Industrial Laser Systems, Electrical Wiring, Soldering, Oscilloscopes, Multimeters, Function Generators, Signal Processing, Laser Diode Drivers, Galvanometer Controls, PLC Programming, HMI Development, Control Panels, AC/DC Converters, Inverters, Transformers, Power Distribution, Industrial Safety Procedures.

Software: Python, C++, C#, MATLAB, Octave, PSpice, Multisim, LabVIEW, SolidWorks, OrCAD.

Professional Experience

LaserStar Technologies – Junior Electrical Engineer (2026–Present)

- Assisted with electrical system design, wiring, troubleshooting, and testing for industrial laser welding and marking systems.
- Supported PCB layout reviews, electrical control systems, and production documentation in compliance with IPC standards.
- Worked with high-voltage power supplies, capacitor charging circuits, laser diode drivers, sensors, motors, and control interfaces.
- Performed bench testing using oscilloscopes, multimeters, function generators, and power supplies to verify system performance.
- Assisted with system integration and communication troubleshooting using RS-232, Ethernet, and CAN bus interfaces.
- Supported engineering change orders (ECOs), cable assemblies, wiring documentation, and production builds.

FedEx Express – Courier / Material Handler (2007–2026)

- Operated in time-critical logistics operations, handling 1,500+ units per shift while supporting on-time aircraft departures.
- Performed customer deliveries, pickups, package handling, scanning, and route coordination.
- Followed safety procedures, equipment handling standards, and operational compliance requirements.

Engineering Projects

- CNC Machine Development: Designed and developed a dual-spindle CNC platform integrating an engraving router and laser module.
- Automated Sorting Machine: Developed a Raspberry Pi and Pi Camera vision system using Python for automated color-based sorting.
- Full-Bridge Inverter Analysis: Performed FFT harmonic analysis and simulation validation using MATLAB and PSpice with 98% agreement.
- Boost Converter Modeling: Modeled converter performance, ripple current, and efficiency across multiple duty cycles using Octave.
- RF Communications System: Built and tested RF communication prototypes using oscilloscopes and spectrum analysis techniques.

Education

B.S. Electrical/Electronics & Computer Engineering – Valencia College, May 2026

A.A. Advanced Electronics Technician – Valencia College, Dec 2024

A.A. Basic Electronics Technician – Valencia College, Dec 2024

Certificate, Automobile Mechanic – Liceo College, San Juan, PR, Aug 2007